
Frisbee Flight

This problem will investigate the flight of a frisbee under the influence of drag.

- (a) The Reynolds number R determines the appropriate drag equation to use for the frisbee's motion. Find the Reynolds number for a frisbee given that $R \propto d$ and depends only on the density ρ of the air, the velocity v of the frisbee, the diameter d of the frisbee, and the viscosity η of the air. Note that the Reynolds number has a numerical prefactor of 1. Let a represent the value of R for $\rho = 10 \text{ kg/m}^3$, $v = 1 \text{ m/s}$, $d = 0.1 \text{ m}$, and $\eta = 10^{-5} \text{ Pa} \cdot \text{s}$.

- (b) The drag force on the frisbee can take two forms:

$$F \propto \eta r v \text{ or } F \propto \rho A v^2$$

A Reynolds number on the order of 10^3 divides the two forms. Find the form of the drag force that is most appropriate for frisbee motion. Let $b = 12345$ if the first form is most appropriate, and let $b = 23456$ if the second form is most appropriate.

- (c) You may assume that the x-component of the frisbee's velocity $v_x \gg v_y$, the y-component of the frisbee's velocity, throughout the motion. You may also assume that the x and y components of the drag force on the frisbee both take the form found above, but with different numerical prefactors k_x and k_y . Given this information, derive an expression for the height of the frisbee above the ground y as a function of the horizontal distance traveled x , the initial speed v_0 , the initial height h , the angle of attack α , k_x , k_y , and any variables given in previous parts.
- (d) Given $v_0 = 14 \text{ m/s}$, $\eta = 1.73 \cdot 10^{-5} \text{ N} \cdot \text{s/m}^2$, $r = 0.15 \text{ m}$, $\rho = 1.29 \text{ kg/m}^3$, $h = 5 \text{ m}$, $A = \pi(0.15)^2 \text{ m}^2$, $m = 0.175 \text{ kg}$, $\alpha = 5^\circ$, $k_x = 0.05$, and $k_y = 0.18$, let c be equal to the sum of the x -coordinates of the frisbee when it is at a height of $y = 6 \text{ m}$ added to the maximum y value attained by the frisbee during its flight added to its x coordinate when it hits the ground ($y = 0$).

Let a , b , and c take their numerical values in SI units. Then, ignoring their units, report the numerical value of $a + b + 1000c$ to the bot. For example, if you found that $a = 10 \text{ s}$, $b = 100 \text{ m}$, and $c = 0.1 \text{ kg}$, you would report $10 + 100 + 100 = 210$ to the bot.

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- (a) By dimensional analysis and noting that the numerical prefactor is 1, $R = \frac{\rho v d}{\eta}$. Thus, $a = 10^5$.
- (b) If we plug in realistic values into our expression for R , we will get that R for a frisbee's motion is on the order of 10^5 . Now, we need to find the form of the drag force that best fits high Reynolds numbers. Notice that when $\eta \ll \rho v d$ the Reynolds number becomes very large. Thus, the second form best fits high Reynolds numbers, as the first form describes a viscous drag force. $b = 23456$.
- (c) Since $v_x \gg v_y$, we see that $v^2 = v_x^2 + v_y^2 \approx v_x^2$. By Newton's laws,

$$a_x = -\frac{k_x \rho A v_x^2}{m}$$

$$a_y = -g + \frac{k_y \rho A v_x^2}{m}$$

We will solve these differential equations to get $x(t)$ and $y(t)$, and then combine the two relations to get $y(x)$.

$$\frac{dv_x}{dt} = -\frac{k_x \rho A v_x^2}{m}$$

$$\int_{v_0 \cos \alpha}^{v_x} \frac{dv_x}{v_x^2} = -\int_0^t \frac{k_x \rho A}{m} dt$$

$$\frac{1}{v_0 \cos \alpha} - \frac{1}{v_x} = \frac{-k_x \rho A t}{m}$$

$$v_x = \frac{dx}{dt} = \frac{v_0 m \cos \alpha}{m + k_x \rho A t v_0 \cos \alpha}$$

$$\int_0^x dx = \int_0^t \frac{v_0 m \cos \alpha}{m + k_x \rho A t v_0 \cos \alpha} dt$$

$$x = \frac{m}{k_x \rho A} \log\left(1 + \frac{k_x \rho A t v_0 \cos \alpha}{m}\right)$$

$$t = (e^{\frac{k_x \rho A x}{m}} - 1) \frac{m}{k_x \rho A v_0 \cos \alpha}$$

Now we find $y(t)$.

$$\frac{dv_y}{dt} = -g + \frac{k_y \rho A v_x^2}{m}$$

$$\int_{v_0 \sin \alpha}^{v_y} dv_y = \int_0^t \left(-g + \frac{k_y \rho A v_x^2}{m}\right) dt$$

$$v_y = \frac{dy}{dt} = v_0 \sin \alpha - gt + \int_0^t \frac{k_y \rho A}{m} \left(\frac{v_0 m \cos \alpha}{m + k_x \rho A t v_0 \cos \alpha}\right)^2 dt =$$

$$v_0 \sin \alpha - gt + \frac{k_y v_0 \cos \alpha}{k_x} - \frac{k_y v_0 m \cos \alpha}{k_x (m + k_x \rho A t v_0 \cos \alpha)}$$

$$\int_h^y dy = \int_0^t (v_0 \sin \alpha - gt + \frac{k_y v_0 \cos \alpha}{k_x} - \frac{k_y v_0 m \cos \alpha}{k_x (m + k_x \rho A t v_0 \cos \alpha)}) dt$$

$$y = h + v_0 \sin(\alpha)t - \frac{1}{2}gt^2 + \frac{k_y v_0 \cos \alpha}{k_x}t - \frac{k_y m}{k_x^2 \rho A} \log(1 + \frac{k_x \rho A t v_0 \cos \alpha}{m})$$

Plugging in our earlier expression for t gives the final answer.

(d) After plotting the graph on Desmos, we read off the values:

$$c = (5.09127 + 35.80878 + 13.65095 + 38.50903) \text{ m} = 93.06003 \text{ m}$$

Our final answer is 216516.03.

Credits:

This problem was inspired by two papers, which can be found here:

[The Physics of Frisbees](#)

[The Aerodynamics of Frisbee Flight](#)

The papers use a numerical computer simulation to graph the trajectory of the flight. This problem goes beyond the analysis of the papers to derive a closed-form expression for the trajectory. I also made an interactive Desmos graph of the trajectory, which can be found [here](#). Please note that my graph differs from the simulated graphs in the papers, as I used slightly different values for some constants. The first paper also contains some errors, which the second paper corrects.